

PROPOSED HL7 FHIR CONNECTATHON 43 TRACK

AI-Assisted Development of Clinical Reasoning Knowledge

Testing how clinical source material can become valid, clinically coherent, interoperable FHIR and CQL knowledge artifacts.

September 19–20, 2026 · Rockville, Maryland

PURPOSE AND INTENT

The track evaluates AI-assisted authoring of computable clinical knowledge

Hypothesis

AI-assisted methods can reduce the time and human effort needed to develop computable clinical knowledge, and can support more artifact types with a repeatable process.

Required result

The resulting knowledge must remain source-traceable, structurally valid, clinically coherent, executable, and portable across independent implementations.

EVALUATION QUESTIONS

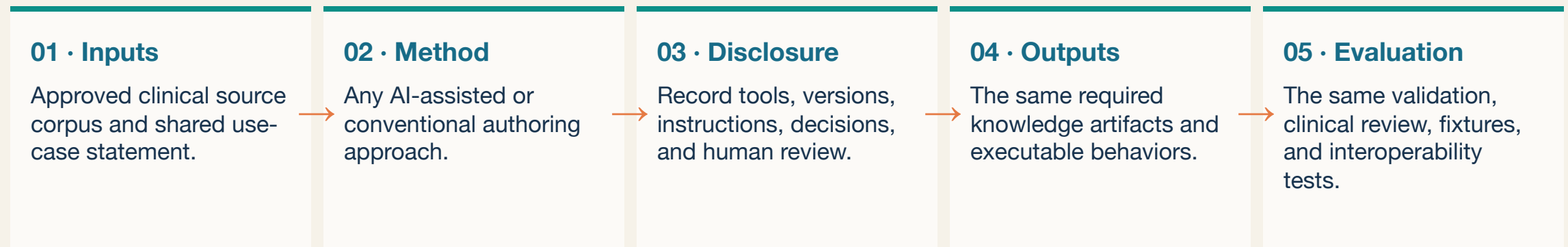
The track records process performance and artifact quality

Dimension	What the track records
Efficiency	elapsed authoring time, human review time, revision cycles, and blocking issues
Scalability	artifact types completed, reuse of the process, and response to source changes
Validity	FHIR validation, canonical resolution, CQL translation, and package checks
Clinical coherence	reviewer findings, corrections, internal consistency, and expected clinical behavior
Interoperability	equivalent results across CQL engines and preservation across FHIR repositories

Where a participant has a comparable conventional baseline, it should be reported. Otherwise, efficiency findings remain descriptive.

COMMON TEST DESIGN

Participants use common inputs and acceptance criteria



No participant implementation defines the reference result. The documented clinical intent and expected behaviors do.

USE CASE

Older-adult fall-risk screening and prevention

Patient-level questions

For a community-dwelling adult age 65 or older:

1. Is the patient in scope for screening?
2. Was the three-question screen completed?
3. Does the result indicate increased fall risk?
4. Which evidence-linked action is applicable?

Population-level question

Across a measurement period, what proportion of eligible patients completed the screen?

Knowledge forms exercised

Evidence · guideline · assessment · measure · terminology · CQL logic

The baseline defines a narrow clinical scope

Included

- community-dwelling adults age 65 or older
- ambulatory context
- CDC STEADI three key questions
- increased risk when any completed answer is yes
- exercise and individualized multifactorial guidance
- track-authored screening-completion measure

Excluded

- diagnosis or emergency triage after a fall
- automatic ordering of an intervention
- inpatient, hospice, or long-term-care populations
- equivalence claims to CMS139FHIR
- real-world quality reporting or clinical deployment

SOURCE MATERIAL

The source corpus covers evidence, recommendation, workflow, and formalization

Source	Role in the use case
CDC STEADI algorithm and pocket guide	three-question assessment and screen–assess–intervene workflow
USPSTF 2024 recommendation	exercise recommendation and individualized multifactorial consideration
USPSTF evidence update and open systematic review	population, outcomes, benefits, harms, and evidence context
LOINC and HL7 CPG / Using CQL with FHIR	terminology and computable-artifact guidance
CMS139FHIR	comparison point only; not copied or represented as the track measure

Redistributable snapshots are included in the repository; standards and comparison references remain link-only. Derived statements must retain stable source locators.

SOURCE-TO-OUTPUT MAPPING

Each source role has a defined computable destination

Source role	Structured knowledge	Primary FHIR output
Evidence review	benefits, harms, population, outcomes	Evidence, EvidenceVariable
Recommendation	applicability, strength, action	PlanDefinition, ActivityDefinition, Library
STEADI workflow	screening and follow-up decisions	PlanDefinition, ActivityDefinition
Three-question screen	questions, answers, completeness, positive result	Questionnaire, Library
Track measure definition	population and completion logic	Measure, Library
Terminology and packaging guidance	bindings, dependencies, canonical identity	ValueSet, ImplementationGuide

EXPECTED OUTPUT

The submission includes reviewable knowledge and FHIR R4 artifacts

Reviewable knowledge

Evidence summary ·
recommendation decision table
· assessment definition ·
measure definition · terminology
definition

FHIR knowledge artifacts

Evidence ·
EvidenceVariable ·
Questionnaire · Library ·
PlanDefinition ·
ActivityDefinition

Measure and package

Measure · ValueSet ·
ImplementationGuide · stable
canonicals · exact dependency
versions

Every clinical statement and operational choice must be distinguishable and traceable to its source or track-authored rationale.

EXPECTED EXECUTABLE BEHAVIOR

CQL expresses assessment, guidance, and measure semantics

Required named behaviors

- In Screening Population
- Completed Three Question Screen
- At Increased Fall Risk
- Exercise Intervention Applicable
- Consider Multifactorial Intervention
- Initial Population, Denominator, Numerator

Deliverable package

- CQL source using a portable subset
- translated ELM JSON and translation logs
- FHIR R4 NPM package
- synthetic patient Bundles
- engine-neutral expected results
- exact translator, engine, and dependency versions

Human review verifies that the artifacts tell the same clinical story

Review questions

- Are source statements separated from track-authored choices?
- Are population, recommendation strength, and actions preserved?
- Do the assessment, guideline, measure, and CQL agree?
- Are uncertainty and incomplete data represented explicitly?

Failure examples

- missing answer treated as a negative answer
- increased risk represented as a diagnosis
- exercise and multifactorial recommendations collapsed
- unsupported code, threshold, or population invented
- measure represented as CMS139FHIR

FHIR validation and CQL compilation are necessary, but they are not clinical approval.

SHARED TEST CASES

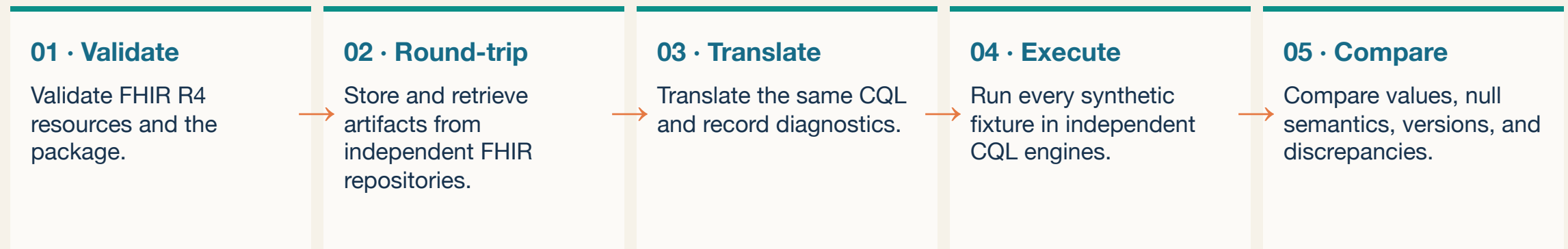
Synthetic fixtures make the expected behavior observable

Case	Expected result
Adult younger than 65	outside screening and measure population
Eligible adult; all answers no	complete screen; not at increased risk
Eligible adult; unsteady yes	increased risk; exercise applicable
Eligible adult; prior fall yes	increased risk; multifactorial consideration
Eligible adult; incomplete response	screen incomplete; risk unknown; numerator false
Eligible adult; no response	screen incomplete; risk unknown; numerator false

Missing data must remain unknown. It must not silently become a negative screen.

INTEROPERABILITY TESTING

Interoperability testing covers validation, translation, execution, and storage



Baseline acceptance requires agreement across at least two independent CQL implementations and round-trip testing in at least two FHIR implementations.

EXPECTED TRACK RESULT

The outcome is evidence about the process and the artifacts

Process evidence

- authoring and review effort
- revision history and human decisions
- repeatability across knowledge-artifact types
- implementation-specific limitations

Knowledge evidence

- reviewed source-to-output traceability
- valid FHIR and translatable CQL
- clinically coherent fixture behavior
- cross-implementation result matrix and issue list

The central question: can AI-assisted development improve efficiency and scale while preserving clinical quality and interoperability?